

1. Administrative & Study Identification

Official Title: A Phase III, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy and Safety of Obinutuzumab in Patients With ISN/RPS 2003 Class III or IV Lupus Nephritis **Study Title:** A Study to Evaluate the Efficacy and Safety of Obinutuzumab in Participants With ISN/RPS 2003 Class III or IV Lupus Nephritis **Short Title/Acronym:** REGENCY **Protocol Number:** CA41705 (2019-004034-42) **NCT Number:** NCT04221477 **Posting ID:** 1018938339259208571 **Sponsor / Company:** Hoffmann-La Roche (Roche) **Investigational Product:** Obinutuzumab (Gazyva / Gazyvaro) **Phase of Development:** PHASE3 (Phase III) **Medical Monitor:** Levi Garraway, MD, PhD (Roche Chief Medical Officer and Head of Global Product Development)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and pharmacokinetics of obinutuzumab compared with placebo in achieving Complete Renal Response (CRR) at Week 76 when added to standard-of-care therapy consisting of mycophenolate mofetil (MMF) and corticosteroids. **Secondary Objectives:** To evaluate the achievement of CRR with reduced corticosteroid use (≤ 7.5 mg/day prednisone), overall safety, pharmacokinetic profile, proteinuric response, mean change in eGFR, and the reduction of disease-causing B cells.

2.2 Background & Rationale Up to one-third of patients with lupus nephritis (LN) go on to develop end-stage kidney disease within 10 years, for which dialysis or transplant are the only available therapeutic options and carry a high risk of mortality. Obinutuzumab is a CD20-directed cytolytic antibody that limits kidney damage by inducing deep B-cell depletion. This study was initiated to address the urgent need for a more effective treatment option by testing whether adding obinutuzumab to standard therapy improves CRR and allows for long-term glucocorticoid exposure reduction.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 **Trial Status:** ACTIVE_NOT_RECRUITING

3.2 Planned Enrollment & Duration Targeted Enrollment: 271 patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 10-Aug-2020 **Estimated Primary Completion:** 15-Aug-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to ≤ 75 years. 2. Diagnosis of active or active/chronic ISN/RPS 2003 Class III or IV proliferative Lupus Nephritis (LN) as evidenced by renal biopsy performed within 6 months. Participants may co-exhibit Class V disease in addition to either Class III or Class IV disease. 3. Urine protein to creatinine ratio greater than or equal to ($\geq$) 1 on a 24-hour collection.

4.2 Exclusion Criteria 1. Pregnancy or breastfeeding. 2. Severe renal impairment or the need for dialysis or renal transplantation. 3. Receipt of an excluded therapy, including any anti-CD20 therapy less than 9 months prior to screening or during screening; or cyclophosphamide, tacrolimus, ciclosporin, or voclosporin during the 2 months prior to screening or during screening. 4. Significant or uncontrolled medical disease which, in the investigator's opinion, would preclude participant participation. 5. Known active infection of any kind or recent major episode of infection. 6. Intolerance or contraindication to study therapies.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Obinutuzumab 1,000 mg on Day 1 and at Weeks 2, 24, 26, and 52 (with or without an additional dose at Week 50). The control group receives a matching placebo. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** 76 weeks for the primary endpoint evaluation (patients without an adequate response may be eligible for open-label obinutuzumab starting at Week 80).

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care therapy consisting of mycophenolate mofetil (MMF) (target dose 2.0 - 2.5 g/day) and corticosteroids (e.g., prednisone tapered to 5 mg/day). **Other Tracked Substances:** Caffeine (Drug Name / Preferred Name: caffeine, ATC Code: N06BC01) is tracked/permitted as a concomitant substance. **Prohibited:** Other immunosuppressive therapies like cyclophosphamide, tacrolimus, ciclosporin, and voclosporin during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Renal Biopsy Confirmation, Eligibility Labs

Baseline | Day 1 | Randomization, First IV Infusion, Standard Therapy Initiation | | **Interim** | Weeks 2, 24, 26, 50, 52 | IV Infusions, Safety Labs, Efficacy Assessment, Corticosteroid Tapering Review | | **End of Study** | Week 76 | Final Primary Endpoint Assessment (CRR evaluation) |

(Note: Additional follow-up and optional open-label extension available post-Week 76)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Percentage of Participants With Complete Renal Response (CRR): Evaluated at Week 76. CRR was defined as an achievement of all the following criteria: urinary protein-to-creatinine ratio (UPCR) <0.5 gram/gram (g/g); estimated glomerular filtration rate (eGFR) $\geq 85\%$ of baseline, as calculated using the chronic kidney disease epidemiology collaboration (CKD-EPI) equation and no occurrence of intercurrent events of rescue therapy, treatment failure, death or early study withdrawal.

7.2 Secondary Outcomes 1. Percentage of Participants Who Achieve CRR With Successful Prednisone Taper (≤ 7.5 mg/day) at Week 76. 2. Percentage of Participants Who Achieve a Proteinuric Response. 3. Mean Change in eGFR. 4. Safety and tolerability profile (AEs and SAEs). 5. Pharmacokinetics and pharmacodynamics.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection of AEs, with a specific focus on infusion-related reactions and infectious complications (e.g., COVID-19-related complications). **Serious Adverse Events (SAEs):** Evaluated throughout the study. The study noted a higher rate of SAEs in the obinutuzumab group vs. placebo (32.4% vs 18.2%), primarily driven by infections, though overall AE proportions were similar. **Data Monitoring Committee (DMC):** Independent safety monitoring panel instituted by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes ($N=271$). Safety Set for adverse event evaluations. **Statistical Methods:** Obinutuzumab and placebo were compared using Cochran-Mantel-Haenszel (CMH) test adjusting for the stratification factors region and race. Percentage have been rounded off. **Handling of Missing Data:** Imputed by multiple imputations using fully conditional specification (FCS) predicted mean matching method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits for verification of source data, focusing on blinded drug administration and response criteria. **Audit Trail:** Continuous eCRF monitoring, data clarification forms, and tracking of protocol adherence.

11. Study Identifiers & Governance

Primary ID: CA41705 **Registry Number (NCT Number):** NCT04221477

Posting ID: 1018938339259208571 **Sponsor/Lead Organization:**

Hoffmann-La Roche (Roche) **Study Title:** A Study to Evaluate the Efficacy and Safety of Obinutuzumab in Participants With ISN/RPS 2003 Class III or IV Lupus Nephritis **Official Regulatory Title:** A Phase III, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy and Safety of Obinutuzumab in Patients With ISN/RPS 2003 Class III or IV Lupus Nephritis

12. Clinical Framework (Lupus Nephritis Specific)

Disease Areas: Lupus Nephritis **Preferred UMLS Name:** Proliferative lupus nephritis **MeSH Code:** D008181 **SNOMED ID:** 68815009 **HPO Codes:** HP:0033726 **Diagnosis Threshold:** Biopsy-confirmed active LN within 6 months **Medical Methodology:** Standard therapy (Mycophenolate Mofetil and Corticosteroids) **Optimization Target:** Complete Renal Response (CRR) at Week 76

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for LN induction and maintenance **Education Protocol (HCP):** Protocol training and standard-of-care tapering guidelines **Education Protocol (Patient):** Informed consent, infusion expectations, and safety monitoring **Quality Audit Mechanism:** Efficacy tracking (CRR) & infection/safety monitoring

14. Observation & Follow-Up Schedule

Total Duration: 76 Follow-up weeks for the primary endpoint analysis **Onsite Visit Cadence:** Day 1, Weeks 2, 24, 26, 50, 52, and 76 **Remote Monitoring Frequency:** Between standard onsite infusion visits **Checklist Review Interval:** Regular safety and dosing visits per protocol

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					